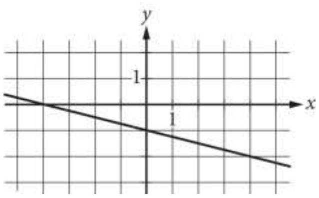
Question ID: b2845d88

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



Which of the following is an equation of the graph shown in the xy-plane above?

Answer

A. $y=-\frac{1}{4}x-1$

B. $y=-x-4$

C. $y=-x-\frac{1}{4}$

D. $y=-4x-1$

Correct Answer: A

Rationale

Choice A is correct. The slope of the line can be found by choosing any two points on the line, such as (4, −2) and (0, −1). Subtracting the y-values results in −2 − (−1) = −1, the change in y. Subtracting the x-values results in 4 − 0 = 4, the change in *x*. Dividing the change in y by the change in *x* yields $-1\div 4=-\frac{1}{4}$, the slope. The line intersects the y-axis at (0, −1), so −1 is the y-coordinate of the y-intercept. This information can be expressed in slope-intercept form as the equation $y=-\frac{1}{4}x-1$.

Choice B is incorrect and may result from incorrectly calculating the slope and then misidentifying the slope as the y-intercept. Choice C is incorrect and may result from misidentifying the slope as the y-intercept. Choice D is incorrect and may result from incorrectly calculating the slope.